Help Desk Active Directory Lab

A home lab I built to practice the Active Directory tasks help desk teams handle every day: password resets, account lockouts, and group management.

Lab Environment

Machine	Role	OS	IP
DC-01	Domain controller, DNS	Windows Server 2022	192.168.10.10
CLIENT-01	Domain-joined workstation	Windows 11 Enterprise (evaluation)	192.168.10.20

- Platform: Oracle VirtualBox
- Network: isolated internal network (`labnetwork`)
- Domain: `lab.local`

What I Built

- Installed and configured Windows Server 2022
- Set a static IP and promoted the server to a domain controller (AD DS + DNS)
- Created an organizational unit, a test user, and a HelpDesk security group
- Built a Windows 11 client, configured its network and DNS, and joined it to the domain
- Verified domain sign-in with a standard user account

Resolved Tickets

Ticket	Summary
Password reset	Reset a user's password and troubleshot a rejected new password
Account lockout	Configured a lockout policy, reproduced a lockout, and unlocked the account

Problems I Solved While Building It

- Fixed a VM that wouldn't boot because the wrong ISO was attached
- Bypassed the Windows 11 TPM/Secure Boot check for a virtual machine
- Fixed a missing network adapter by changing the virtual adapter type

- Resolved memory limits on an 8 GB laptop by running the VMs in separate sessions

Skills Practiced

Active Directory Users and Computers, DNS, Group Policy, static IP configuration, domain join, user account management, troubleshooting, documentation

Planned Next

- Group membership changes
- Group Policy objects (desktop restrictions)